



Hochiminh City University of Technology
Computer Science and Engineering
[CO1027] - Fundamentals of C++ Programming

Course Introduction

Lecturer: Phuoc Hung Pham
Duc Dung Nguyen
Credits: 3

Introduction

- Audience: students who have **no background** in computer programming
- Aims: provide basic knowledge and skill on programming with two important programming paradigms: **structure programming** and **object-oriented programming**.
- Demonstration language: C++
- Prerequisite: basic math knowledge
- Requirement:
 - Class attendance
 - Self-study
 - Work hard

Learning outcome

- What you will get from the course
 - Be able to describe the algorithm for your problem
 - Understand and be able to use structure programming techniques
 - Be able to implement a given algorithm using C++
 - Understand basic concepts of Object-Oriented Programming (OOP)
 - Improve your coding style
 - The process of solving problem

Contents

- Basic of programming language (C++)
- Control structures
- Array and structure
- Pointer
- Recursive
- Class
 - Inherirance, template, polymorphism, and advanced topics

Syllabus

- Course meeting time:
 - Lecture: 3 hours/week for 8 weeks
 - Laboratory: 2 hours/week for 9 weeks
- Course mechanics:
 - Textbook: C++ How to program
 - Reference book: *Fundamentals of C++ Programming* – Richard L. Halterman
 - Lecture notes
 - Online materials

Syllabus

- Assessment
 - Assignment
 - Lab test
 - Final exam: 90'
 - Ratio: lab (10%), test (20%), assignment (30%), final exam (40%)
(This is a tentative ratio, it may change a bit)
- Coding environment:
 - Recommend: Visual studio
 - Other IDEs are welcome

Regulations

- ❖ Any plagiarism act will lead to zero in all tests!
- ❖ Final grade of assignment depends on the exam
 - ❖ $A_{\text{final}} = N \sum_{i=1}^N \frac{1}{T_i^{-1}}$
- ❖ Detail mapping of exam questions and assignments will be announced during the progress of the course.